

Paulo Eduardo dos Santos

SENIOR DATA ENGINEER | CLOUD & BIG DATA | WEB3 & FINTECH DATA

ATS Match: 100% — Senior Data Engineer @ Consensys

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PROFESSIONAL SUMMARY

Senior Data Engineer with 9 years building and operating large-scale data pipelines and platforms for banking, payments, and e-commerce organizations processing millions of daily transactions. Strong in Python and SQL with production experience across cloud data warehouses (Snowflake, BigQuery, Redshift), transformation and orchestration with dbt and Airflow, and big-data processing with Apache Spark, AWS EMR, and S3. Embeds data quality, data governance, and security (PII/LGPD) into every pipeline, deploys infrastructure as code with Terraform, and automates CI/CD. Builder of public open-source blockchain-data projects (Ethereum, Solana, DeFi) — combining enterprise data engineering with hands-on Web3.

TECHNICAL SKILLS

Languages: Python (advanced), SQL (advanced), PySpark, Shell Script, Scala

Transformation & Orchestration: dbt, Apache Airflow, ETL/ELT design, CDC, Kafka

Cloud Warehouses: Snowflake, BigQuery, Amazon Redshift

Big Data & Distributed Processing: Apache Spark, AWS EMR / EMR Serverless, S3, Athena, Trino, Hive

Data Modeling: dimensional modeling (star schema, Kimball), data warehouse design, Medallion/lakehouse (Delta Lake, Apache Iceberg)

Data Quality & Governance: data quality frameworks, data catalog (AWS Glue, Unity Catalog), PII/LGPD compliance, HashiCorp Vault, AWS KMS

Cloud Platforms: AWS (S3, EMR, Glue, Athena, Redshift, Lambda, Kinesis, Lake Formation, Step Functions, KMS), Azure (Databricks, Data Factory, Synapse), GCP (BigQuery, Dataproc)

Infrastructure & CI/CD: Terraform, Docker, Kubernetes, Git, CI/CD

Web3 / Blockchain Data: on-chain analytics (Ethereum, Solana), DeFi/NFT metrics, Dune Analytics

Visualization: Power BI, QlikSense, Tableau, Grafana

OPEN-SOURCE PROJECTS (GITHUB.COM/PAULOEDUSPBR)

- **dune-analytics-queries** — 12 SQL queries on Dune analyzing blockchain data: DeFi protocol metrics, NFT markets, Ethereum and Solana activity, cross-chain comparisons.
 - **crypto-data-pipeline** — production-style pipeline ingesting market data for 100 crypto assets (CoinGecko API) with Airflow, PostgreSQL, and Docker; 7/7 automated data quality checks passing.
 - **medallion-data-lake** — production-ready Bronze/Silver/Gold template with Airflow, dbt, and Terraform (AWS + GCP), SCD Type 2 dimensions and 36 automated tests.
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PROFESSIONAL EXPERIENCE

Data Engineer — BRQ

Jul 2025 – Present | Remote, Brazil

RD Saúde — Healthcare Data Lake (Customer 360)

- Built a healthcare Data Lake with Medallion architecture (Bronze/Silver/Gold) on Apache Iceberg over S3, supporting electronic health records, scheduling, vaccines, medication, prescriptions, and tele-consultation use cases (Amplimed, HAOC, Vitat).
- Developed PySpark/SparkSQL pipelines for large-scale ingestion, transformation, and consolidation of clinical data, with support for Apache Iceberg and Apache Hudi.
- Orchestrated Airflow DAGs integrated with Spark on Kubernetes, with automated YAML manifest generation

through Python builders.

- Modeled the Customer 360 analytical table integrating multiple sources (CRM, scheduling, events, interests) served via Trino, Athena, and Redshift.
- Implemented an in-house Data Quality framework (cross-origin validations, light checks, DLS validations) and a YAML-versioned data catalog.
- Secured sensitive health data (PII/LGPD) with HashiCorp Vault, AWS KMS, and Fernet encryption, including decrypt pipelines in landing.
- Integrated API ingestion (Amplimed, HAOC, Precifica) with async S3 clients and evolving schema handling (JSON/Parquet/Iceberg).
- Built end-to-end observability (custom log handlers, Spark metrics, exception tracking, Iceberg table comments) integrated with AWS SNS and Step Functions; maintained Docker containers (dev/qa/prod) for PySpark and Python K8s jobs.

Itaú — Modernization & Analytics

- Developed ETL pipelines using EMR Serverless and Glue Studio; optimized SQL queries for performance and implemented data quality processes.
- Performed A/B test analysis and developed metric-improvement analyses and hypothesis validation.

Senior Big Data Specialist / Data Engineer — Stefanini

Jun 2023 – Jun 2025 | São Paulo, Brazil

Banco do Brasil — Big Data Environment Sustainment

- Sustained and optimized a production Cloudera/Hortonworks cluster, ensuring **99.9% uptime SLA**.
- Developed Python and Shell automation for operational processes; built Jupyter notebooks for exploratory analysis and Ansible jobs for infrastructure management.
- Optimized Hive queries and maintained HDFS structures; built dbt models in Snowflake for financial reporting with data quality tests and versioned transformations.

AmericaNet — APIs & Data Processing

- Developed a RESTful API in Python Flask to expose processed data; implemented complex transformations with SparkSQL in Databricks.
- Automated processes with Azure Automation and built data pipelines with error handling and logging; built dbt/Snowflake ELT models with integrity validations and Unity Catalog.

Cielo — Dashboard Analytics

- Analyzed and documented legacy Oracle procedures; developed PySpark scripts for high-volume processing and Athena external tables over S3 with partitioning.
- Migrated analytical data to Snowflake with dbt dimensional modeling supporting Power BI dashboards.

Itaú — ETL Process Refactoring

- Refactored legacy SQL tables to Parquet and converted stored procedures to PySpark jobs, achieving **60% storage reduction**.
- Built a data catalog with AWS Glue, implemented data quality processes, and produced detailed technical documentation of all migrated processes.

DirectLog — Geolocation System

- Built a pipeline capturing latitude/longitude via the Google Maps API and Airflow DAGs for batch address processing.
- Implemented an authenticated SFTP web interface, EC2 hosting, and a caching layer to optimize API calls.

Brasildev — AWS Data Lake

- Built a layered Data Lake (Bronze/Silver/Gold) with a data catalog and governance; created a Python ingestion engine with Salesforce integration.
- Configured security/access policies with Lake Formation and automated infrastructure with Terraform.

Senior Data Engineer — Natura & Co

Feb 2022 – Jun 2023 | São Paulo, Brazil

E-commerce Pipeline

- Modeled dimensional data for e-commerce analytics; developed ETL pipelines with EMR Serverless and Airflow

DAGs for orchestration.

- Built Delta Lake structures for data versioning and incremental load processes; integrated with BI via Redshift.
- Built dbt models in Snowflake for customer-behavior analysis with data quality tests.

Cielo — Cloud Migration & Modernization

- Analyzed Oracle data structures and mapped them to BigQuery; migrated legacy data to Azure Blob Storage and built pipelines in Azure Data Factory with Unity Catalog cataloging.
- Converted complex SQL logic to distributed processing; developed PySpark and SparkSQL on Databricks.

JBS — Data Pipeline Modernization

- Provided technical leadership for a team of engineers and developers.
- Implemented a high-availability NiFi cluster integrated with Active Directory; configured an Application Load Balancer and built NiFi flows for SAP ingestion and S3↔Redshift movement.

Data Engineer — Enel

May 2021 – Feb 2022 | Brazil

Data Warehouse Migration

- Built a migration pipeline with Talend Open Studio moving data from Impala to Redshift.
- Created QlikSense dashboards connected to Redshift, documented existing ETL processes, optimized queries, and trained business teams.

Big Data Solutions Architect — Banco do Nordeste

Jan 2021 – May 2021 | Brazil

Corporate Data Lake

- Designed a Data Lake architecture with raw, stage, and analytics zones; implemented an unmanaged Hadoop cluster and configured Dremio for the analytics layer.
- Planned a NoSQL structure with Cassandra for a credit system; integrated legacy systems via DataStage.

Bradesco — Data Ingestion Framework

- Updated ingestion-framework scripts for Hive and HBase tables; developed shell automation and integrated with the IBM IWS scheduler; connected to Teradata for extraction.

Big Data Architect / Engineer — Hospital Sírio-Libanês

Mar 2019 – Dec 2020 | São Paulo, Brazil

Hospital Analytics Platform

- Analyzed and ingested hospital data from multiple sources; installed and configured Airflow for orchestration and developed ETL DAGs.
- Implemented Docker containers integrated with Airflow, configured Kubernetes for scalability, integrated AWS Cognito with Azure AD for SSO, and set up a CloudWatch→Elasticsearch pipeline via Lambda.

Raízen — Data Ingestion Framework

- Surveyed the data-pipeline architecture and analyzed Azure Data Lake, Databricks, and Power BI; documented SAP integration via SSIS Theobald.

Dell EMC (Client: Fleury) — Consulting & Training

- Sustained a Hortonworks environment; configured security with Kerberos/Ranger and AD integration; developed Oozie monitoring scripts, built Grafana dashboards, and trained the client team.

Big Data Technical Lead — Claro Brasil

Sep 2017 – Mar 2019 | São Paulo, Brazil

Cloudera Cluster Implementation

- Installed and configured 3 Cloudera clusters (Data Lake, M2M, BI); implemented Kerberos with AD as KDC and cross-realm trust; integrated Hive/Impala with BI tools (SAS, QlikView, Alteryx).
- Built a data ingestion framework and monitoring/sustainment scripts and integrated Informatica BDM for ETL — cutting processing time by **50%**. Mentored and led 2 engineers.

Nextel — Sustainment & ETL

- Developed ETL ingestion processes, analyzed data with Beeline/Impala, extracted data via Sqoop, gathered

requirements, and sustained the production environment.

Big Data Analyst — Vivo

Jun 2017 – Sep 2017 | Brazil

Hadoop Migration & Sustainment

- Built ETL ingestion processes, analyzed data with Hive and Python, migrated Hadoop 2.2 → 2.6, persisted data in HBase, and managed the cluster with Zookeeper.
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EDUCATION

- **Associate Degree in Big Data and Data Intelligence** — Anhembi Morumbi University | Expected 2026
 - **Artificial Intelligence** (in progress) — FMU
 - **Bachelor of Science in Information Systems** — UNIP | 2006
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CERTIFICATIONS

- Red Hat Certified System Administrator (RHCSA)
 - Big Data Engineering — Garre Training (160h)
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LANGUAGES

- **Portuguese:** Native
- **English:** Upper-Intermediate (B2) — fluent technical reading and writing